By the way...
I gave up on my ideas for helicopters for 20 years when my early attempts were unsuccessful.

How it changed the world

The amazing maneuverability of helicopters means they can do many things that planes can't, making them ideal for difficult rescue missions, especially on mountains and at sea. They can also do some pretty amazing stunts!

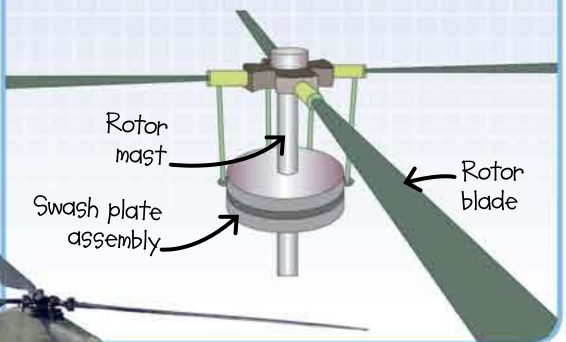
Take off!

The first **practical helicopter** got off the ground in the early 1930s, but it was Russian-American Igor Sikorsky's VS-300 that today's helicopters are based on. It used a **LARGE ROTOR** on top for lift, and a tail rotor to keep it steady. It first flew in 1939, and was soon wowing onlookers. Its novel design meant that it ***could move in almost any direction*** (even upside down), and hover.

Sikorsky later designed the VS-44, a flying boat for passengers.

HOW IT WORKS

A helicopter's main rotor blades provide lift. The pilot can move the aircraft up, down, backward, and forward by changing the rotors' speed and angle (via the swash plate assembly) in relation to the wind. Hovering happens when the lift from the rotor equals the pull of gravity. The tail rotor stops the helicopter from spinning, and controls the left and right movement of the craft.



It paved the way for...

The **HOVERCRAFT**, *which also uses high-pressure air to create lift, was developed by Englishman Christopher Cockerell in 1956.*

Twin-rotor helicopters, such as **CHINOOKS**, *were invented in the 1960s by American Frank Piasecki.*